Automation in Construction 132 (2021) 103935

Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon

Deep learning-based road damage detection and classification for multiple countries

Deeksha Arya [a,b,*], Hiroya Maeda [b], Sanjay Kumar Ghosh [a,c], Durga Toshniwal [a,d], Alexander Mraz [b,e], Takehiro Kashiyama [b], Yoshihide Sekimoto [b]

[a] Centre for Transportation Systems (CTRANS), Indian Institute of Technology Roorkee, 247667, India
[b] Institute of Industrial Science, University of Tokyo, 4-6-1 Komaba, Tokyo, Japan
[c] Department of Civil Engineering, Indian Institute of Technology Roorkee, 247667, India
[d] Department of Computer Science and Engineering, Indian Institute of Technology Roorkee, 247667, India
[e] Amazon EU, 22 Rue Edward Steichen, 2540, Luxembourg

ARTICLE INFO

Keywords:
Automatic road condition monitoring
Convolutional neural network
Deep learning
Smartphone-based road damage detection
Big Data
Smart infrastructure

ABSTRACT

Many municipalities and road authorities seek to implement automated evaluation of road damage. However, they often lack technology, know-how, and funds to afford state-of-the-art equipment for data collection and analysis of road damages. Although some countries, like Japan, have developed less expensive and readily available Smartphone-based methods for automatic road condition monitoring, other countries still struggle to find efficient solutions. This work makes the following contributions in this context. Firstly, it assesses usability of Japanese model for other countries. Secondly, it proposes a large-scale heterogeneous road damage dataset comprising 26,620 images collected from multiple countries (India, Japan, and the Czech Republic) using smartphones. Thirdly, it proposes models capable of detecting and classifying road damages in more than one country. Lastly, the study provides recommendations for readers, local agencies, and municipalities of other countries when one other country publishes its data and model for automatic road damage detection and classification. A part of the proposed dataset was utilized for Global Road Damage Detection Challenge'2020 and can be accessed at (https://github.com/sekilab/RoadDamageDetector/).

1. Introduction

Road infrastructure is a crucial public asset as it contributes to economic development and growth while bringing critical social benefits. It connects communities and businesses and provides access to education, employment, social, and health services. However, road surface wears and deteriorates over time from factors related to location, age, traffic volume, weather, engineering solutions, and materials used to construct it and, therefore, the knowledge of its deterioration extent are critical to efficient and cost-effective maintenance with the goal to preserve its good and safe condition. The impact of roadway deficiencies on safety and economy has been widely studied [1,2].

Pavement distress, one of the pavement condition characteristics, can be evaluated using three approaches: manual, semi-automated, or fully automated. The traditional methods for collecting pavement condition data involve manual and semi-automated surveys.

In manual approach, raters perform a visual inspection of the pavement surface through either walking or driving along the pavement surface using a slow-moving vehicle. Visual inspection of the road surface suffers from a subjective judgment of inspectors. It requires a significant human intervention that is proven to be time-consuming, given the extensive length of road networks. Moreover, inspectors must often be physically present in the travel lane, exposing themselves to potentially hazardous conditions.

In semi-automated pavement condition evaluations [3–5], the road images are collected automatically from a fast-moving vehicle, but distress identification is postponed to an off-line process running in workstations at the office. This approach improves safety but still uses manual distress identification, which is very time-consuming.

Fully automated distress evaluations often employ vehicles equipped with sophisticated and expensive sensors. The collected data in the form of images is then processed using pattern recognition software for

Abbreviations: API, Application Programming Interface; CNN, Convolutional Neural Network; RDD, Road Damage Dataset; SSD, Single Shot Detector.

* Corresponding author at: Centre for Transportation Systems (CTRANS), Indian Institute of Technology Roorkee, 247667, India.
E-mail address: deeksha@ct.iitr.ac.in (D. Arya).

https://doi.org/10.1016/j.autcon.2021.103935
Received 30 November 2020; Received in revised form 31 July 2021; Accepted 30 August 2021
Available online 11 September 2021
0926-5805/© 2021 Elsevier B.V. All rights reserved.

identifying road distress. The data processing may be accomplished during data collection or later as post-processing at the office. Raters' role is to conduct quality assurance testing of the software functionality along with quality control of the software's output for distress rating.

Specialized vehicles used for automatic pavement inspection are usually equipped with multiple sensors such as laser scanners, road profilers, and cameras with the aim to capture road assets, including pavement images, and to acquire the longitudinal and transverse profiles of the road (for instance, refer [6–8]). However, such vehicles are expensive, and the cost to purchase such systems can reach half a million dollars, depending on the sensors included. Even though the operating price of road survey vehicles is cheaper than that of traditional surveying methods, it can reach between $30 and $50 per kilometer [9].

Meanwhile, mobile devices such as smartphones have evolved in recent years into devices containing high-resolution digital cameras, sensors, and powerful processors; therefore, examples of road inspection using smartphones are becoming more common. The advantage of employing smartphones is that they enable efficient and cost-effective inspection of the road surface of large road networks. For example, Mertz et al.[10] proposed a method to handle road images acquired by on-board smartphones installed on cars that operate daily, such as general passenger automobiles, buses, and garbage trucks, to detect road surface damage with an external laptop. Casas-Avellaneda and López-Parra [11] proposed a method to visualize potholes detected by smartphone sensors on a map. Maeda et al.[12] developed a smartphone application for the collection and real-time detection of roadway deficiencies for Japanese roads.

Even though Japanese municipalities have started using the Smartphone-based model, there are many municipalities and local authorities in several other countries still lacking the technology, know-how, and funds to afford expensive state-of-the-art data collection equipment. On many occasions, they sustain their road networks through inefficient maintenance planning, and in some circumstances, they are unable to carry out adequate inspections.

In the context of this challenging situation regarding the maintenance of such infrastructure, there is a need to assess the usability of the Japanese road damage detection and classification model for monitoring the road conditions in other countries. Further, it needs to be analyzed whether there would be any effect on the accuracy level of the Japanese model when used for other countries. If yes, what are the options available for other countries if the performance of the Japanese model is not found satisfactory when applied to their local roads? Do they need to train a new model for themselves from scratch, or can they utilize the Japanese model or data in some other way?

The questions raised above need to be addressed based on some detailed research. Further, the recommendations are often sought for readers of other countries when one country, in this case, Japan, publishes its data or model for a specific application.

The presented work intends to answer the aforementioned research questions and provide the readers with apt recommendations regarding the automation of road damage detection and classification.

For the required comprehensive analysis, this study considers two countries, namely India and the Czech Republic (partially Slovakia), and conducts several experiments using different combinations of underlying data.

Firstly, the localized road damage datasets were created by using 3590 road images from Czech and 9892 images from India using smartphones installed on the windshield of the vehicle. The newly collected datasets were labeled for cracks and potholes present in the images and were mixed with the Japanese dataset.

Thereafter, the study involves training and evaluating 16 deep neural network models considering 30 scenarios based on different combinations of the test and train datasets for detecting and classifying road damages. The evaluation results of these models are analysed to provide recommendations for the readers of other countries.

Overall, the contributions of this work can be listed as follows:

1. Performance analysis of the usability of the Japanese model for detecting and classifying the road damages in other countries.
2. Proposing a large-scale heterogeneous dataset, comprising 26,620 annotated road images collected using a vehicle-mounted Smartphone from multiple countries.
3. Proposing Deep Learning-based models capable of detecting and classifying road damages in more than one country.
4. Providing recommendations for readers, local agencies and municipalities of other countries when one country publishes its data and model for automatic road damage detection and classification.

The following section provides an overview of the existing literature.

2. Related work

2.1. Machine learning and convolutional neural networks

Deep Learning which build upon neural networks, is an advanced concept of Machine Learning and provides solutions in application domains that are difficult to model with conventional statistical approaches [13].

Convolutional Neural Network (CNN) is one of the main categories of deep neural networks that are usually applied for image recognition and classification [14]. These are the type of algorithms that can identify street signs, cars, faces, and many other types of objects. The main advantage of CNN is that after the training stage, it automatically detects the critical features without any human supervision.

Many CNN architectures, such as R-CNN (Region-based Convolutional Neural Networks [15]), Fast-CNN [16], or Faster-CNN [17] have been developed to attain the best accuracy while improving the processing speed. However, the computational load was still too large for processing images on devices with limited computation, power, and space [18]. Therefore, the Single Shot Multi-Box Detector(SSD) framework was developed to improve further the computation speed of the object detection [19]. It uses a single feed-forward convolutional network to detect multiple objects within the image directly and combines predictions from numerous feature maps with different resolutions to handle objects of various sizes.

MobileNet is a small, low-latency, and low-power convolutional feature extractor that can be built to perform classification, detection, or segmentation similar to popular large-scale models, such as Inception SSD [20]. It is based on depth-wise separable convolution, which factorizes a standard convolution into a depth-wise convolution, and 1×1 convolution called a point-wise convolution. Further, Huang et al. [21] compared many neural networks and object detection methods, and among them, the SSD object detection framework using a MobileNet base network was found to require relatively small CPU load and memory consumption while maintaining high accuracy.

Thus, the MobileNet is designed to effectively maximize accuracy while being mindful of the restricted resources for on-device or embedded applications. It has also been shown to achieve accuracy comparable to VGG-16 on the ImageNet dataset with only $1/30^{th}$ of the computational cost and model size [22]. Due to all these advantages, the study presented in this paper adopted SSD MobileNet for training the deep neural network models using road images.

2.2. Machine learning-based road infrastructure inspection

Recent studies have adopted various Machine Learning and Deep Learning-based approaches for automated road surface survey or damage detection [23–25]. For instance, Supervised deep convolutional neural network was trained by Zhang et al., [26] to classify pavement images taken by smartphones around the Temple University campus. The proposed road damage detection methods focused on the determination of the existence of damage. Similarly, Silva and Lucena [27] developed a machine learning-based model to detect cracks on concrete

Table 1
A summary of the existing road image datasets and research works along with the proposed work.

Dataset	Study area	Images	Resolution	Data collection device	Acquisition process		View captured	
					Stationary	Dedicated vehicle	Top-down	Wide-view
Ouma and Hahn [31]	Eldoret Town, Kenya	75	1080 × 1920	Samsung Galaxy S5 camera	×	×	✓	×
CrackIT [32]	_	84	1536 × 2048	Optical device	_	_	✓	×
CFD [33,34]	Beijing, China	118	480 × 320	iPhone 5 camera	✓	×	✓	×
CrackTree200 [35]	_	206	800 × 600	Area-array camera	_	_	✓	×
Weng et al. [36]	Highway G45 in China	217	2048 × 1536	_	_	_	✓	×
SDNET2018 [37]	Logan, Utah, USA	230	4068 × 3456	16MP Nikon Digital Camera	_	_	✓	×
Li et al. [38]	Northern China	400	_	Automatic Road Analyzer	_	_	✓	×
Crack500 [39]	Philadelphia, USA	500	2000 × 1500	Cell Phone	_	_	✓	×
EdmCrack600 [40]	Edmonton, Canada	600	1920 × 1080	GoPro 7	×	×	×	✓
GAPs v1 [6]	Germany	1969	1920 × 1080	Professional Camera, Mobile mapping system	×	✓	✓	×
GAPs v2 [24]	Germany	2468	1920 × 1080	Professional Camera, Mobile mapping system	×	✓	✓	×
Majidifard et al. [41]	United States	7237	640 × 640	Google API			✓	✓
Maeda et al. [12]	Japan	9053	600 × 600	LG Nexus 5X Smartphone	×	×	×	✓
Angulo et al. [42]	Japan, Italy, Mexico	18034	600 × 600	Smartphone	×	×	×	✓
Proposed Work (RDD-2020)	India, Japan, Czech Republic	26,620	600 × 600, 720 × 960	LG Nexus 5X, Samsung Galaxy J6 Smartphone	×	×	×	✓

Table 2
Further details for the research work and datasets listed in Table 1.

Dataset	Damage category		Annotation-level			Techniques implemented
	Cracks	Potholes	Pixel	Block/ Boundary-box	Image	
Ouma and Hahn [31]	×	✓	✓	×	×	Wavelets filtering, fuzzy c-means, Morphological reconstruction
CrackIT [32]	✓	×	✓	✓	×	k-nearest neighbor, Parzen windowing, etc.
CFD [33,34]	✓	×	✓	×	×	Random Structured Forests and Random Decision Forests
CrackTree200 [35]	✓	×	✓	×	×	Geodesic shadow-removal, Minimum Spanning Tree
Weng et al. [36]	✓	×	✓	×	×	Structured Edge Detector, Morphological Operation etc.
SDNET2018 [37]	✓	×	×	×	✓	AlexNet DCNN architecture
Li et al. [38]	✓	×	×	×	✓	Back Propagation Neural Network
Crack500 [39]	✓	×	✓	×	×	Feature pyramid and hierarchical boosting network
EdmCrack600 [40]	✓	×	✓	×	×	ConnCrack combining conditional Wasserstein GAN and connectivity maps
GAPs v1 [6]	✓	✓	×	×	✓	ASINVOS net
GAPs v2 [24]	✓	✓	×	✓	×	ASINVOS net and ResNet34
Majidifard et al. [41]	✓	✓	×	✓	×	YOLOv2 and Faster-RCNN
Maeda et al. [12]	✓	✓	×	✓	×	SSD Inception-v2, MobileNet
Angulo et al. [42]	✓	✓	×	✓	×	RetinaNet
Proposed Work (RDD-2020)	✓	✓	×	✓	×	MobileNet (and other techniques as part of GRDDC([46])

surfaces. Further, Anand, Gupta, Darbari, and Kohli [28] presented the development of deep neural network architecture using GPU board and associated camera to detect road cracks and potholes for self-driving cars and autonomous robots to perform necessary evading maneuvers to ensure a smooth journey. Likewise, Fan et al., [29] trained Supervised convolutional neural network (CNN) to recognize different pavement conditions, and results presented in the paper show that it can deal well with varying pavement textures. Furthermore, the authors Nhat-Duc, Nguyen, and Tran [30] shows superiority of deep neural network-based approach as compared to edge-detection-based method for recognizing pavement cracks.

Recently, Karaaslan, Bagci, and Catbas [43] proposed a semi-supervised deep learning-based method to analyse infrastructure damage targeting cracks and spalls. The authors propose an approach based on attention guidance. The proposed approach detects the boundary box for the damage and get it verified by the human-inspector, after which the pixel-level segmentation is applied inside the detected bounding box. The results show that the approach significantly reduces the computational cost of the segmentation since only the region of interest is used unlike other comparable models (such as MaskRCNN) which apply segmentation operation on the entire image and carry out localization in parallel.

In another work, the authors Guan et al.[8] integrated stereo-vision and deep-learning and performed segmentation-based analysis for cracks and potholes. The authors created Multi-feature image datasets comprising 2D, 3D as well as enhanced-3D images using stereo vision. The authors proposed the use of a modified U-net based on depth-wise separable convolution, for faster segmentation. Further, the 3D image segmentation was performed to automatically measure the volume of the potholes. Moreover, some researchers focused on detecting road damage using other type of sensor data instead of using images. For instance, the authors in [44] employed accelerometer sensors to record the vehicle vibration while driving and used machine learning to assess the road pavement conditions based on the collected data. Likewise, Singh et al.[45] collected the data using the inbuilt sensors of the smartphones (accelerometer and Global Positioning System) and used Dynamic Time Warping (DTW) technique for detecting road surface conditions. The results presented in the study show that the technique DTW worked faster than the machine learning techniques Support Vector Machines and Artificial Neural Networks.

Nonetheless, the current study focuses mainly on two dimensional-image-based approaches for automatic road damage detection, some of which are summarized and compared with the proposed work in Tables 1 and 2.

Clearly, the research works listed in the table differ in several factors, including number of images used, the way of data collection, type of models implemented etc. A significant factor in deciding the approach or network architecture considered by the researchers is the type of

annotation used for data preparation. Mostly, the pixel-level annotations lead to designing segmentation-based approaches for road damage detection. For block-level, object detection-based approaches are generally preferred. Image-level annotation is used by researchers with objective to only detect the presence or absence of the damage in the image.

While a majority of researchers focus on a single type of annotation and designing corresponding algorithms, some researchers consider multiple aspects. For instance, Oliveira and Correia[32] considered both block-based and pixel-based detection of cracks and proposed a toolkit CrackIT using several image-processing and machine learning-based techniques. The pixel-based mode of CrackIT provides good crack localization but is sensitive to noise. The block-based mode is not so good for crack localization, but can reject the noise efficiently. The authors Abdellatif et al.[47] considered the aforementioned pros and cons and extended the work by combining the block-based and pixel-based approaches. The combined method outperforms the pixel-based mode as well as three other deep learning-based methods proposed in [48,49,39].

Thus, the results indicate that the performance of pixel-based approaches can be improved by combining them with block-based approaches. The current work considers the block-based approach that involves the road damage annotation using boundary boxes.

As specified earlier, the proposed dataset is an extended version of the existing Japanese Road Damage dataset [12,50]. The following subsection discusses the existing literature related to the same.

2.2.1. Literature related to the Japanese road damage dataset and model

The detection method developed by Maeda et al.[12] classified pavement deterioration of Japan road network into eight categories based on images captured by mobile devices. The corresponding dataset, named RDD-2018, was made publicly available in 2018, and a Smartphone-based application was also introduced for real-time road condition assessment. Since then, this application is being used by several municipalities in Japan for faster monitoring of road conditions. As a result, the underlying data, method, and models have gained wide attention from researchers all over the world.

A technical challenge was organized in December 2018 as a part of the IEEE Big Data Conference held at Seattle, USA, which utilized this data for evaluating the performance of several models for road condition monitoring. In total, 59 teams participated in the challenge from 14 different countries. These teams, although provided solutions having better accuracy than the original models included in [12], the new solutions were mostly based on changing the underlying network models and hyper-parameter configurations [51–54]. Although some authors suggested improvements in the dataset but the teams themselves did not modify the dataset in their work.

After that, some researchers focused on experimenting with the underlying dataset by either adding more images to it or by introducing a completely new dataset based on a similar method utilized by Maeda et al. [12]. For instance, Angulo et al. [42] extended this dataset by adding images collected from Italy and Mexico. The size of the dataset was increased to 18,034 images, and the authors mostly focused on having a more balanced representation for individual damage classes, especially potholes.

Similarly, Roberts et al.[55] used the data collection app introduced by Maeda et al. [12] and collected over 7000 road images from Sicily, Italy. Their work utilizes only the newly collected images, but the underlying methodology is similar to [12], except that they also consider severity analysis for the identified road damages.

The presented study differs from these works in the following ways:

1. The Smartphone-based dataset introduced in this work is larger and more heterogeneous, covering multiple countries.
2. These works do not consider the perspective of evaluating the responses of existing or newly proposed models for detecting damages in different countries individually.
3. The Japanese dataset has been updated by re-annotating and adding more images from Japan, named RDD-2019, in recent work [50]. Unlike the previous studies based on the Japanese dataset (2018 version) or models, the presented work builds upon this updated dataset.
4. The presented work also analyze the effect of varying the number of images in training dataset on the performance of the trained models. This also lays the foundation for analyzing whether a model trained on data from a single country or multiple countries performs better.

Table 3
Road damage types and definitions considered in Maeda et al. [12].

Damage type			Detail	Class name
Crack	Linear Crack	Longitudinal	Wheel-marked part	D00
			Construction joint part	D01
		Lateral	Equal interval	D10
			Construction joint part	D11
	Alligator Crack		Partial pavement, overall pavement	D20
Other Damage			Pothole	D40
			Cross walk blur	D43
			White line blur	D44

2.2.2. Other similar works

The authors in [56] uses a dataset of 45,788 road images collected from Shanghai and utilize YOLO network for detecting and classifying pavement distresses. However, the images are collected using an industrial high-resolution camera, unlike the presented work, which is based on the lesser expensive Smartphone-based images.

Likewise, Majidifard et al.[41] use Google street view images considering both top-down as well as wide-view for characterization and density calculation of pavement distresses collected from 22 different pavement sections in the United States. The results indicate that the YOLO v2 model outperformed the Faster R-CNN model in distress classification. The free availability and easy accessibility of Google Street view images is the major attraction of the work. However, only 7237 images were used in this work despite considering nine different categories of the damage. Similar to the research by Majidifard et al. [41], the authors in [57] proposed a CNN-based predictive model to detect potholes using Google API images. Nevertheless, the analysis carried out by the authors is based on only 3424 images (including both pothole and non-pothole categories).

Ideally, more than 5000 labeled images are generally required for each class for an image processing based classification task to provide satisfactorily accurate results [13]. Although the Google Street view images are easily and freely available, the annotating the images is still a time-taking and laborious task. This further escalates the need to analyze whether the data already available from another country can be mixed with the local data for improving the performance of a road damage detection and classification model.

This study intends to address the research gaps and provides the recommendations required for practical use of the models by different countries.

The following section provides detail of the proposed dataset and the methodology followed by this study to design the experiments and carry out the requisite analysis.

3. Datasets and methodology

This section first explains the transition from the dataset RDD-2018

to the proposed dataset, followed by an explanation of the dataset used in this study.

3.1. Road damage dataset 2018

RDD-2018 was introduced and made publicly available in 2018 [12], and since then, it has been used exhaustively by several researchers to propose new methods for automatic condition monitoring of roads as well as other infrastructures. Table 3 shows the damage categories and the corresponding definitions used in this dataset.

3.2. Road damage dataset 2019

Maeda et al. [50] reviewed and relabelled the contents of the dataset RDD-2018 and added new annotated images to form the new dataset RDD-2019. The number of images with annotations were increased from 9053 to 13,135. The number of annotations increased from 15,435 to 30,989.

3.3. The proposed road damage dataset 2020

The proposed dataset builds upon the recently introduced RDD-2019 dataset. This is to be noted that the road damage data included in Road Damage Dataset 2018 and 2019 comprise road images collected from only a single country, that is, Japan. However, the dataset proposed in this study considers data collected from multiple countries. The significant points of difference are as follows.

1. The total number of images is increased to 26,620, almost thrice the prevailing 2018 dataset.
2. New images were collected from India and Czech Republic (partially from Slovakia) to make the data more heterogeneous and train robust algorithms.
3. Unlike previous versions, this dataset considers only four damage categories, comprising mainly of cracks and potholes, namely D00, D10, D20, and D40. Note that the standards related to evaluations of Road Marking deterioration such as Crosswalk or White Line Blur differ significantly across several countries. Thus, these categories were excluded from the study so that generalized models can be trained applicable for monitoring road conditions in more than one country.

More details related to the newly collected images from India and the Czech Republic are provided as follows. It can be noted that a part of the proposed RDD-2020 dataset was utilized for Global Road Damage Detection Challenge 2020. The statistical distribution for the same is provided in our companion paper [46] with additional information in the corresponding data article [58].

3.3.1. Study area

The details of study area for road images from Japan are provided in Section 3.1 of the article [12]. Additionally, road images for the presented study were collected from India, the Czech Republic and partially, in Slovakia. For the Czech Republic and Slovakia, a large portion of road images was collected in Olomouc, Prague, and Bratislava municipalities and included a mix of 1st class, 2nd class, 3rd class roads, and local roads. A smaller portion of the road image dataset was collected along D1, D2, and D46 motorways to improve the robustness of the trained model. Similarly for India, a mixture of images collected from local roads, state highways, and national highways was considered, covering the metropolitan (Delhi, Gurugram) as well as non-metropolitan regions (mainly from the state Haryana). All these images were collected from plain regions. Road selection and time of data collection were made based on road accessibility, weather conditions, and traffic volume.

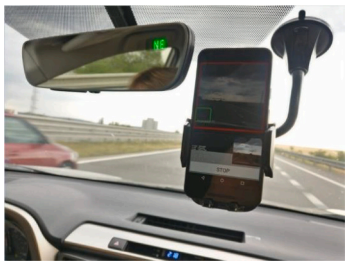


Fig. 1. Installation setup of the smartphone in the car.

3.3.2. Data collection

Road images were captured using a smartphone running a publicly available image-capturing application developed by Sekimoto Lab, The University of Tokyo[12]. The smartphone was installed on the dashboard/windshield of the vehicle, and the application captured JPEG images with a resolution of 600 × 600 pixels at a rate of one image per second (Fig. 1). For India, the updated version of the application was used, and the images with a resolution of 720 × 960 pixels were captured. These images were resized to 720 × 720 to have square images similar to Japan and Czech to maintain uniformity in the dataset for different countries. The aforementioned image capturing rate was selected to enable the collection of pictures without overlap or leakage when the vehicle traveled at an average speed of approximately 40 km/h (or 25 mph). However, vehicle speed on motorways and some 1st class roads/highways outside the cities was adjusted to comply with the local laws. Further, it can be noted that the proposed data majorly capture the images of flexible(asphalt) pavements.

For the Czech Republic and Slovakia, a total of 10,400 images were collected between March and October 2018 under varying weather and lighting conditions, including sunny, overcast, light rain, and sunset. However, only 3595 images were found suitable for the study. The remaining images which were either blurred or were not covering the required road damage categories or significant portions of the road were discarded. Similarly, for India, a total of 10,595 images were captured in October 2019, out of which, 9892 images were finally selected.

3.3.3. Data classification/damage categories

Based on the Japanese Maintenance Guidebook for Road Pavement [59], Maeda et al. [12] use eight damage categories in total. This classification covers two types of deterioration, namely pavement deterioration (D00, D01, D10, D11, D20, D40) and road marking deterioration (D43, D44), as shown in Table 3. The sample images are shown in Figure 2 in the article [12]. However, the standards for evaluating the road pavement conditions are not the same across the countries and vary significantly. For instance, the Czech Catalog of Deficiencies in Flexible Pavements, Czech Ministry of Transport [60] recognizes 28 different pavement deterioration divided into three main categories, including loss of asphalt mixture, cracks, and deformations. This catalog does not recognize Crosswalk marking deterioration as a type of pavement deficiency. Similarly, for India, the code IRC:82 [61] provides guidelines for the maintenance of bituminous/flexible pavement in India. It identifies four main types of pavement distresses, namely, Cracking, Deformation (rutting, upheavals, etc.), Disintegration (raveling, potholes, edge breaking, etc.), and Surface Defects that relate to inapt quality and quantity of bitumen. Since the aim of this study is to consider the perspective of monitoring road conditions in more than one country, it includes only the four major categories of damage mainly comprising cracks and potholes, which are generally common to almost all the countries. Thus, for this study, the damage categories considered are - D00 to represent Longitudinal/ Parallel cracks, D10 for Transverse/Perpendicular cracks, D20 for Alligator/Complex cracks, and D40 for Potholes. Some other categories which are indistinguishable in the images from the above mentioned four cases, for instance, rutting is

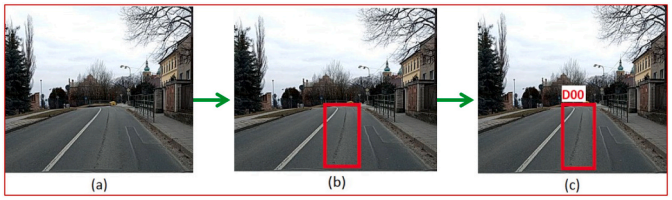


Fig. 2. Annotation pipeline: (a) original image, (b) image with the bounding box, (c) final annotated image containing bounding box and class label.

marked as D40 in Japanese dataset [12], are covered along with these so as to provide a generalized model. Further, this is to be noted that the existing Japanese datasets (RDD-2018 and 2019) also contain annotations for damage categories (D01, D11, D43, D44, D50), which are not included in this study. This work makes no changes to these already available annotated images. Instead, it uses algorithms that read only the annotations related to the desired four categories.

3.3.4. Data annotation

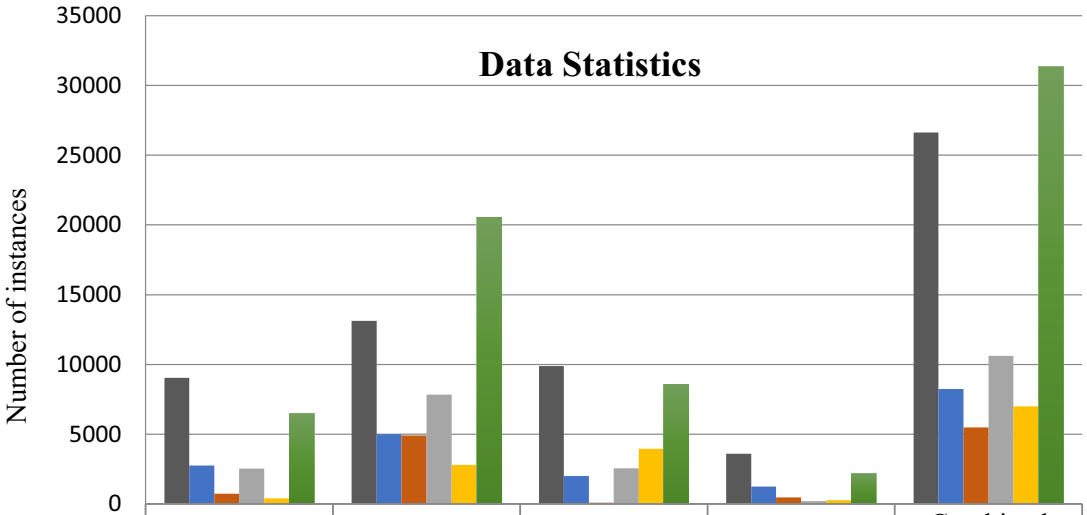
Data annotation is a fundamental part of what makes many machine learning projects function accurately. It provides the initial setup for teaching a deep neural network to recognize objects and discriminate them against various input images. In this work, the collected road images were manually annotated using labelImg software. The annotation pipeline is presented in Fig. 2. Annotation is a labor-intensive procedure and took considerable time because the complete dataset was first manually reviewed on the screen and then all recognized deficiencies were annotated by enclosing them with bounding boxes and classified by attaching the proper class label. Class labels and bounding box coordinates, defined by four decimal numbers (x_{min}, y_{min}, x_{max}, y_{max}), were stored in the XML format similar to PASCAL VOC [62]. Next, the data was converted into TFRecord file format, as required by the TensorFlow Object Detection API [63].

3.3.5. Statistics

Fig. 3 shows the statistics for the proposed dataset. A comparatively lesser number of images are used from the countries India and Czech than what was available from Japan in RDD-2019. This is because motive of this study is to analyze and provide solutions in which the dataset already available from other countries can be utilized to design models for new countries, such that the requirement for collecting new local data is minimized. Further, this can be noted that the number of instances for different damage categories in Indian and Czech datasets is

Table 4
Naming convention for the models.

Model name	Training data (#images)			Tested for
	Japan	India	Czech	
Ja_2k	2000			Japan, India, Czech
Ja_4k	4000			
Ja_6k	6000			
Ja_8k	8000			
Ja_10k	10,000			
In_2k		2000		India
In_4k		4000		
In_6k		6000		
In_8k		8000		
Cz_2k			2000	Czech
InJa_12k	10,000	2000		India &Japan
InJa_14k	10,000	4000		
InJa_16k	10,000	6000		
InJa_18k	10,000	8000		
JaCz_11k	10,000		1000	Czech
JaCz_12k	10,000		2000	



	Japan (2018)	Japan (2019)	India	Czech	Combined (InJaCz)
Total #images	9053	13133	9892	3595	26620
D00	2768	5006	1994	1242	8242
D10	742	4911	89	480	5480
D20	2541	7846	2556	211	10613
D40	409	2798	3947	263	7008
Total #labels	6460	20561	8586	2196	31343

Fig. 3. Statistics for the number of damage instances included in the underlying datasets.

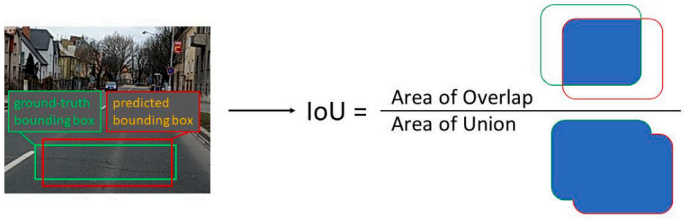


Fig. 4. An illustration for calculating Intersection over Union (IoU).

a bit dis-balanced. This leads to the requirement of analyzing if the data from another country can be used to create a balanced representation. For example, only 89 instances of transverse cracks, that is, D10 are present in the 9892 images available from India, which is very less. Combining it with dataset available from Japan may help in representing this class better while training the requisite neural network models. Similarly, for alligator cracks, D20, and potholes, D40, in the case of Czech data, adding data from another country or collecting new local data becomes a necessity for training efficient models. The following section illustrates the methodology followed in this study.

3.4. Experimental setup and the methodology

The study involves training and evaluating 16 deep neural network models considering 30 scenarios based on different combinations of the test and train datasets, as described in Table 4. The critical differences in the models are:

1. Source of the training data: Following two cases were considered:
   (a) Single Source Models - It covers the models trained on a single source of data (Japan) applied to test data from different countries (India, Japan, Czech).
   (b) Multiple Source Models - It covers the following two categories:

Table 5
F1-score for the experiments.

Experiment	Model Name	Test Data(1000 images)	Damage Category			
			D00	D10	D20	D40
E-1	Ja_2k	Japan	0.196	0.084	0.404	0.305
E-2	Ja_4k		0.257	0.138	0.450	0.305
E-3	Ja_6k		0.268	0.154	0.421	0.356
E-4	Ja_8k		0.293	0.194	0.435	0.334
E-5	Ja_10k		0.308	**0.228**	0.504	0.401
E-6	Ja_2k	India	0.024	_	0.050	0.015
E-7	Ja_4k		_	_	0.022	0.048
E-8	Ja_6k		0.030	_	0.034	0.010
E-9	Ja_8k		0.032	_	0.022	0.052
E-10	Ja_10k		0.051	_	0.028	0.036
E-11	Ja_2k	Czech	0.110	0.062	0.064	_
E-12	Ja_4k		0.156	0.082	0.019	_
E-13	Ja_6k		0.158	0.022	0.046	_
E-14	Ja_8k		0.120	0.109	0.045	_
E-15	Ja_10k		0.175	0.097	0.022	_
E-16	In_2k	India	0.141	_	0.432	0.266
E-17	In_4k		0.140	_	0.448	0.295
E-18	In_6k		0.199	_	**0.502**	**0.333**
E-19	In_8k		**0.226**	_	0.442	0.307
E-20	Cz_2k	Czech	0.263	0.071	**0.282**	**0.196**
E-21	InJa_12k	India	0.159	_	0.325	0.276
E-22	InJa_14k		0.129	_	0.391	0.257
E-23	InJa_16k		0.200	_	0.403	0.311
E-24	InJa_18k		0.224	_	0.487	0.305
E-25	InJa_12k	Japan	0.327	0.174	0.522	0.369
E-26	InJa_14k		**0.361**	0.177	0.502	**0.434**
E-27	InJa_16k		0.343	0.111	0.512	0.366
E-28	InJa_18k		0.335	0.128	**0.524**	0.336
E-29	JaCz_11k	Czech	**0.324**	**0.177**	0.204	0.076
E-30	JaCz_12k		0.318	0.128	0.228	0.092

Table 6
mAP (mean Average Precision) for the models considered in the study for test datasets of varying size.

Experiment	Model Name	Test Data	AP(Damage Category)				mAP
			D00	D10	D20	D40	
E-1	Ja_2k	Japan	4.29%	1.46%	19.14%	11.76%	9.16%
E-2	Ja_4k		7.41%	3.63%	20.70%	11.23%	10.74%
E-3	Ja_6k		8.32%	3.49%	18.39%	14.71%	11.23%
E-4	Ja_8k		9.93%	6.09%	21.47%	13.00%	12.62%
E-5	Ja_10k		9.72%	**6.28%**	27.29%	17.09%	15.10%
E-6	Ja_2k	India	0.09%	0.00%	1.42%	0.46%	0.49%
E-7	Ja_4k		0.00%	0.00%	0.50%	1.11%	0.40%
E-8	Ja_6k		0.12%	0.00%	0.57%	0.25%	0.24%
E-9	Ja_8k		0.19%	0.00%	0.44%	1.03%	0.42%
E-10	Ja_10k		0.51%	0.00%	0.36%	0.56%	0.36%
E-11	Ja_2k	Czech	2.09%	0.65%	0.61%	0.00%	0.84%
E-12	Ja_4k		4.81%	1.49%	0.10%	0.00%	1.60%
E-13	Ja_6k		2.73%	0.26%	0.43%	0.00%	0.86%
E-14	Ja_8k		1.76%	2.14%	0.27%	0.00%	1.04%
E-15	Ja_10k		4.98%	1.97%	0.19%	0.00%	1.79%
E-16	In_2k	India	2.40%	0.00%	21.42%	7.62%	7.86%
E-17	In_4k		3.04%	0.00%	21.68%	9.45%	8.54%
E-18	In_6k		4.16%	0.00%	**26.27%**	**11.29%**	10.43%
E-19	In_8k		**5.84%**	0.00%	21.45%	10.06%	9.34%
E-20	Cz_2k	Czech	7.04%	2.06%	**11.59%**	**4.46%**	**6.29%**
E-21	InJa_12k	India	2.89%	0.00%	11.99%	7.86%	5.69%
E-22	InJa_14k		1.86%	0.00%	17.21%	6.59%	6.42%
E-23	InJa_16k		4.36%	0.00%	17.43%	11.46%	8.31%
E-24	InJa_18k		5.52%	0.00%	25.86%	10.52%	**10.48%**
E-25	InJa_12k	Japan	10.81%	4.37%	28.00%	15.25%	14.61%
E-26	InJa_14k		**13.20%**	4.35%	25.68%	**19.57%**	**15.70%**
E-27	InJa_16k		12.53%	2.87%	28.40%	16.39%	15.05%
E-28	InJa_18k		12.12%	3.81%	**29.35%**	15.44%	15.18%
E-29	JaCz_11k	Czech	**10.14%**	**5.23%**	4.25%	1.35%	5.24%
E-30	JaCz_12k		10.06%	2.33%	8.51%	1.64%	5.64%

i. Pure Modelling or Same Source, Same Target Modelling: A model trained and tested on data collected from the same country.
ii. Mixed Modelling: It includes mixing the local data of some other country, generally the target country, with publicly available data from Japan for training the models.

2. Size of the training data: The number of images considered for training the models was varied from 2000 to 18,000 based on the availability of the data from different countries. This was done to analyze the effect of varying the size of training data on the performance of the models.

Further, since the images available from different sources were originally in different resolutions as described in data collection, images were re-scaled to a resolution of 300 × 300 pixels while training the models. TensorFlow API 1.80 and official Python distribution 3.6.5 were used to construct, train, and deploy the object detection models.

The study utilizes the SSD MobileNet model pre-trained on the MSCOCO (Microsoft Common Objects in Context) dataset [64] to train the requisite road damage detection models using transfer-learning. The last layer of the base model MobileNet was modified to suit the current application and Weight Freezing was used for the remaining layers. Further, Mini-batch gradient descent algorithm was used with a batch size of 24 for training the models. The learning rate was set to 0.003, and RMSprop optimizer (introduced in 2012 by T. Tieleman and G. Hinton in a COURSERA lecture, Neural Networks for Machine Learning), with momentum optimizer value = 0.9 was applied. All models were trained for 250,000 steps. Training epochs can be calculated using these steps and the size of training dataset, using the Eq. (1).

$$\text{Epochs} = \frac{(\text{Batch Size}) \times (\text{Training Steps})}{(\text{Size of Training Dataset})} \quad (1)$$

4. Results and analysis

4.1. Evaluation parameters

The performance of the trained model was estimated by using three indices, namely, Precision, Recall, and F1-score. Precision indicates the percentage of correctly predicted features (i.e., true positives) out of the total number of predicted features (i.e., true and false positives). The Recall shows the percentage of correctly predicted features out of the total number of features present in the actual class (i.e., true positives and false negatives).

Precision and Recall are both based on evaluating Intersection over Union (IoU), which is defined as the division of the area overlap between predicted and ground-truth bounding boxes by the area of their union, as shown in Fig. 4. Because Precision and Recall counter each other and increasing one of them will usually reduce the other, The most common way to reach a balance between these metrics is to use F1-score. It

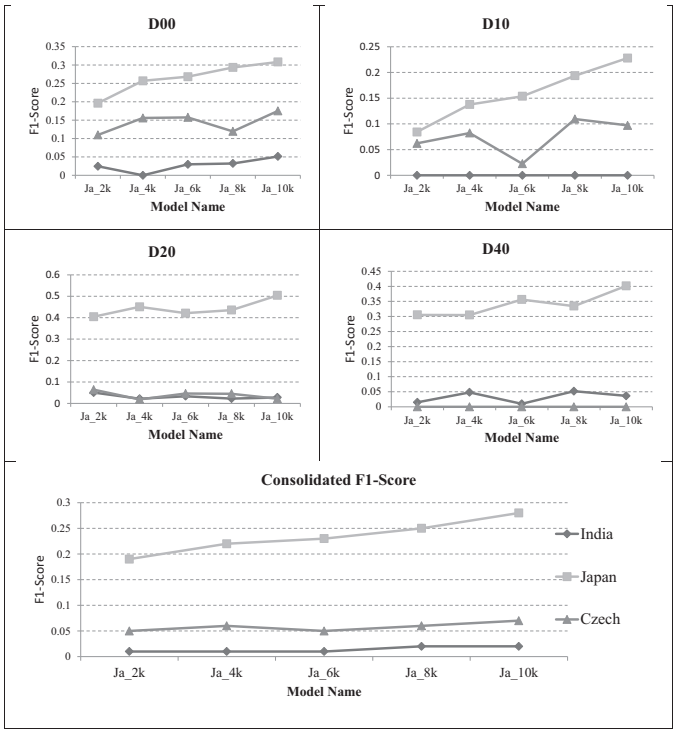


Fig. 5. F1-score for Models trained using Japanese data and applied to test data from different countries.

measures the overall model's accuracy and is calculated as:

$$F_1 = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \tag{2}$$

Maximizing the F1-score ensures reasonably high Precision and Recall.

Additionally, mean Average Precision (mAP) is used for the quantitative analysis. The mAP for a model is defined as mean of Average Precision (AP) for all damage categories. The Average Precision is calculated as the area under the Precision and Recall Curve for each damage category, that provides insights of the performance over the complete range of Recall values.

For evaluation of the model's performance, the IoU threshold was set to 0.5, which represents an evaluation metric prescribed by the PASCAL VOC object detection competition [62].

4.2. Evaluation results

Table 5 presents the value of the F1-score obtained for different damage categories for all the experiments conducted in the study. The highlighted (bold) entries show the maximum F1-score obtained for each country for different damage categories. Next, Table 6 shows the values of Mean Average Precision obtained for different models when applied to test datasets of varying size and location. The empirical analysis for these results is presented in the following subsection.

4.3. Empirical analysis

4.3.1. Single source modeling

Fig. 5 shows the F1-score results for different damage categories when a model trained on Japanese data is applied to different target countries by using test data constituting 1000 images from Japan, India, and Czech, respectively.

As depicted by the individual and the consolidated chart, the model trained on Japanese data, though worked fine for Japanese test data; the performance is abysmal when applied to test data from India and Czech,

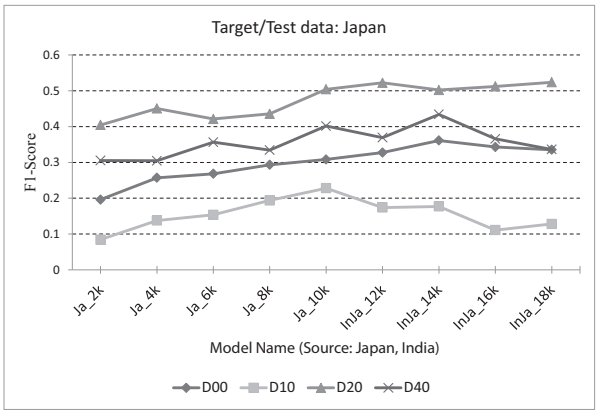


Fig. 6. Model name vs. F1-score (Target: Japan).

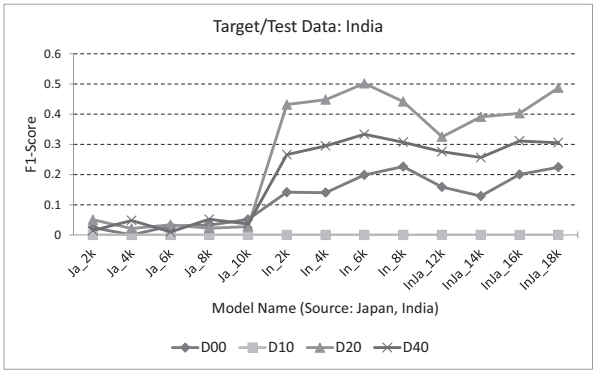


Fig. 7. Model name vs. F1-score (Target: India).

irrespective of the number of images used for training. Though, the model performs better for some damage categories than others, the overall performance is very low.

4.3.2. Multiple source modeling

The study analyzes the models trained using multiple sources of data based on the target or the test data, as illustrated below.

4.3.2.1. Target: Japan. Fig. 6 shows the F1-score results for various models applied to Japanese test data. Clearly, the models having training data as a mixture of images from India and Japan show better performance for Japan for D00, D20, and D40. For D10, the performance shows a downfall on increasing the number of images added from India because the number of instances for the D10 category is almost negligible in Indian data (89 in total). And, thus, as the total number of images increases on adding Indian data to Japanese data, the proportion of D10 instances in the total number of damages decreases, leading to the decline in the performance of the final trained model in identifying D10.

4.3.2.2. Target: India. Fig. 7 shows the compiled results for different models applied to test data from India. The results illustrate that a model trained using Indian data (either pure or mixed in the form of InJa models) perform significantly better than the model trained using only Japanese data. Notably, even the models trained using Indian data fail to perform satisfactorily for Transverse Cracks (D10) in Indian data. This is because only 89 instances of transverse cracks are present in the 9892 images available from India, leading to the presence of only 9 to 10 images in Indian test data constituting 1000 images in total. These 10 images are not sufficient for evaluating the performance of the object detection models leading to NaN(Not a Number, shown by _ in the table) values in F1-score. The NaN for F1-score marks the cases when both

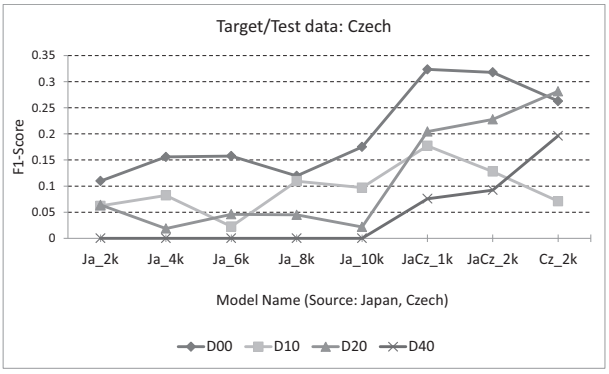


Fig. 8. Model name vs. F1-score(Target: Czech).

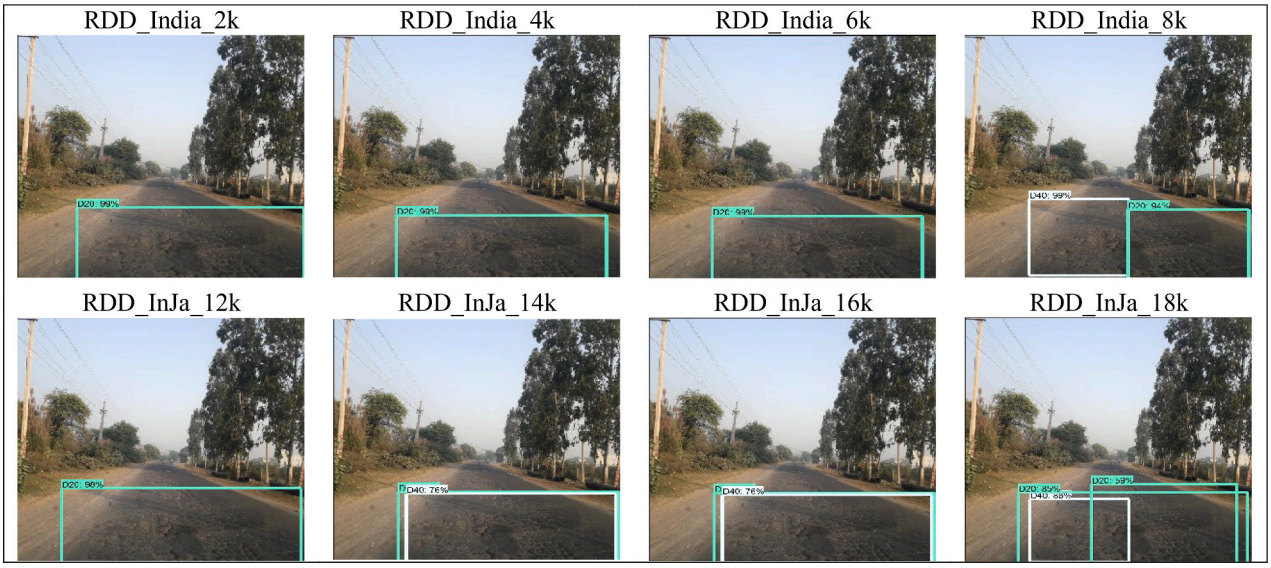


Fig. 9. Labels predicted for sample image 1 by models trained using RDD_India and RDD_InJa.

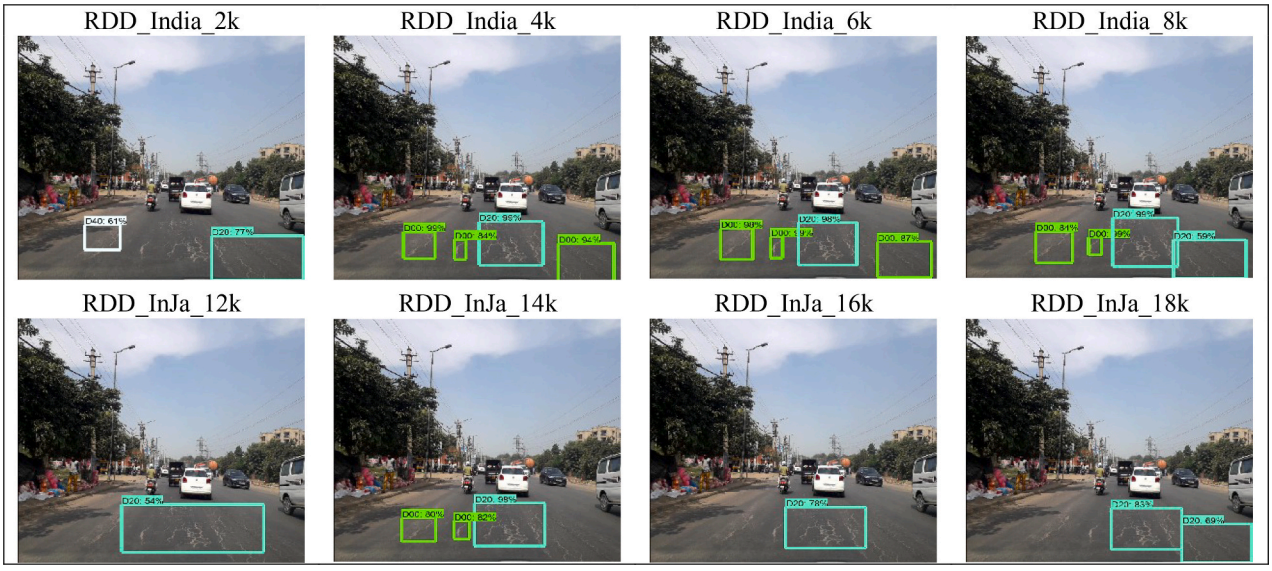


Fig. 10. Labels predicted for sample image 2 by models trained using RDD_India and RDD_InJa.

Precision and Recall values are obtained as zero. For other categories, the observed trend shows adding images from Japan does not help much in improving the performance of the models trained using only Indian data. Nonetheless, the performance of pure Indian models is slightly better than the mixed InJa models, despite having smaller training datasets.

4.3.2.3. Target: Czech. Fig. 8 shows the results for different models applied to the test data from the Czech. Analyzing the graph from right to left shows, a model trained using only 2000 images of Czech (Cz_2k) shows better average performance than the model trained using Japanese data constituting up to 10,000 images. However, this may be a case of overfiting. Further, mixing the images from Czech with the already available Japanese data (10,000 images) shows some improvement over the case when using data from a single country. Two cases were considered for the analysis, based on the number of images used from Czech. The observations are listed as follows:

1. Adding images from Japan to Czech data helps in improving the performance for Linear cracks, both longitudinal as well as transverse.
2. For Alligator cracks and Potholes - the pure Czech model seems to outperform the mixed JaCz models. This can be explained as follows. The number of instances for D20 and D40 in Czech data is very less, and mixing a huge amount of Japanese data for training the model decreases the proportion of Czech D20 and D40 instances further. This leads to the dominated effect of corresponding features learned from Japanese data and ultimately to degraded performance of JaCz models for identifying D20 and D40 of Czech.

4.4. Visual analysis

As described in the analysis based on F1-score, the behavior of models trained for Japan or Czech is easily understandable. However, the models trained using only the Indian datasets show comparable performance to the models trained using the Indian dataset mixed with 10,000 Japanese images, which is a bit unusual. Usually, the performance should get improved on increasing the size of the datasets by such a large extent. To better understand this uncommon behavior of the models, the corresponding prediction results are analyzed visually. The ground truth for the images used for visual analysis is provided in the Appendix.

Figs. 9 and 10 show the labels predicted by several models trained using Indian (In_2k, In_4k, In_6k, In_8k) and combined Indian-Japanese (InJa_12k, InJa_14k, InJa_16k, InJa_18k) datasets for different Indian road images. The image in Fig. 9 shows a high severity alligator crack intermixed with pothole-like structures. For the given image, India_8k identified that the alligator cracks have taken the form of potholes. InJa_14k could identify the presence of D40 with D20, though not individually. InJa_18k provided the most precise result, even though the model detected two overlapping boundary boxes for D20.

Clearly, there cannot be a single correct label boundary for this type of image. The labels predicted by all the models are acceptable when a comprehensive survey for monitoring road conditions is required. However, since the evaluation of object detection models involves comparing the predicted labels with the information in the ground truth file, some of the predicted labels, although acceptable, get marked as false examples. When many such images are present in the dataset, it results in low performance of some of the models. This may be a reason for the low accuracy of InJa viz.a.viz. Indian models, despite having larger training datasets. The features learned by InJa models are a bit dominated by the Japanese dataset since the number of images used from Japan is more than that of India in all the cases.

For the image in Fig. 10, the outputs of Indian models are too much specific, providing a hint that these models may be over-fitting the Indian data. For InJa models, InJa_14k shows better accuracy than the remaining models. To analyze if the Indian models are over-fitting the Indian data, the study analyze the behavior of these models for data collected from Japan and Czech.

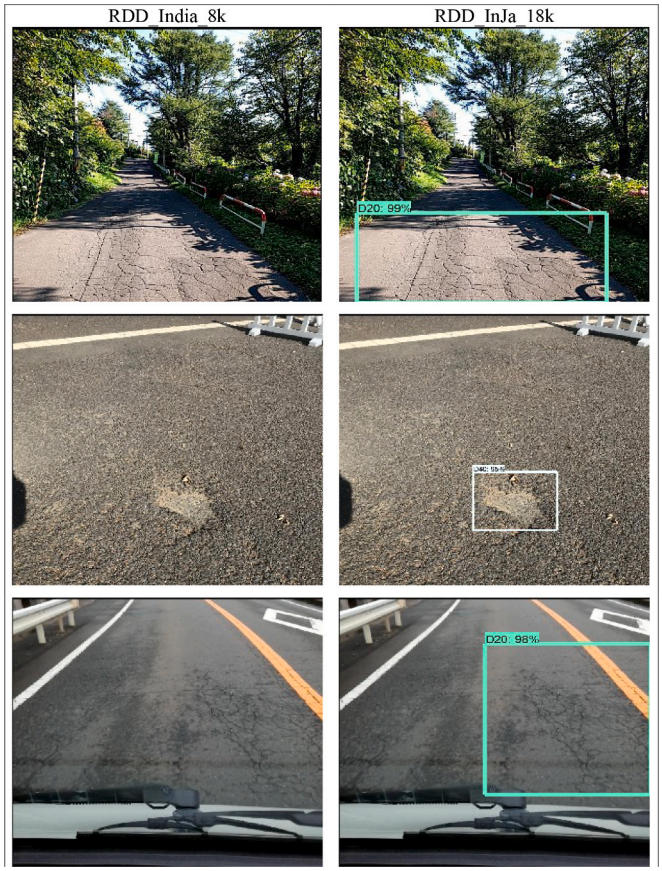


Fig. 11. Sample images from Japan and Czech for comparing the performances of In_8k and InJa_18k models (cases of missed detection by In_8k).

Fig. 11 shows some sample images from Japan and Czech in which In_8k missed some damages, but InJa_18k identified them correctly. Fig. 12 shows the images for false detection by Indian or InJa models. From top to bottom, the first image demonstrates the falsely detected D20 on the roadside by the InJa model. However, the pattern is relatable to D20. For the second image, the pothole-like structures were missed entirely by In_8k and mis-classified by InJa_18k. Thus, in case the motive of the survey is to detect the presence or absence of the damages, the performance of InJa_18k can be counted as better than In_8k. The last image shows the case of false-positive detection by India_8k.

In summary, it can be said that although the models trained using only Indian data show performance comparable to models trained by mixing Japanese data with Indian datasets (InJa), InJa models are more generalized. Also, for InJa models, InJa_14k is found to have the best overall performance, combining the evaluations for both India and Japan, and can be used efficiently for detecting and classifying road damages in both the countries.

4.5. Observations

The performance analysis based on F1-score shows that mixed source modeling outperforms single-source as well as pure modeling in the case of multiple sources. Further, the following observations can be made:

1. Adding data from other countries helps in increasing the generalizing ability of the models, as demonstrated by the performance of several Indian, Japanese, and mixed India-Japan models.

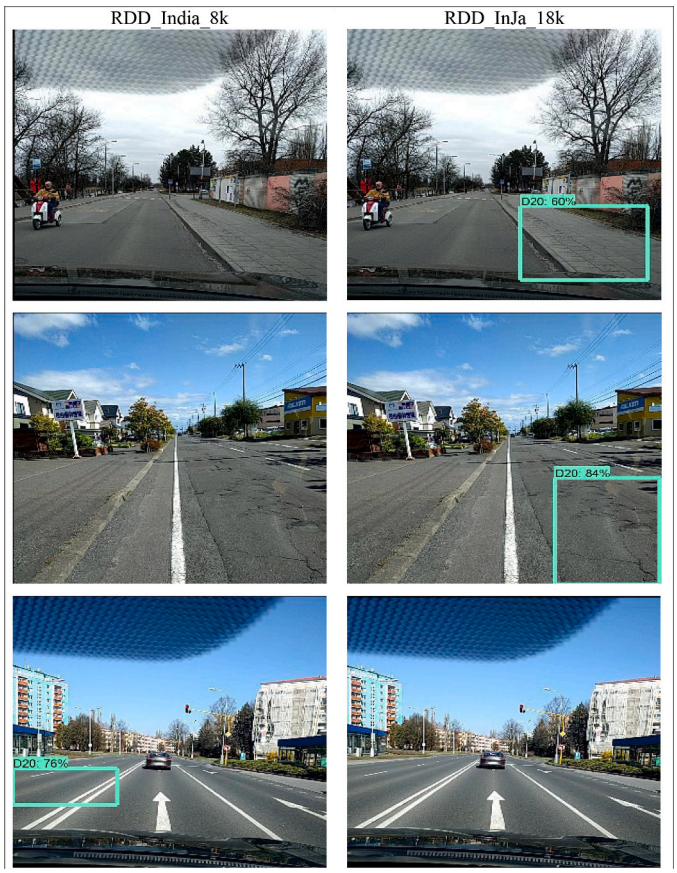


Fig. 12. Sample images from Japan and Czech for comparing the performances of In_8k and InJa_18k models (cases of false detection).

2. The effect on improving the performance by adding data from another country may not be the same/mutual/bidirectional for two countries. For instance, mixing Indian and Japanese data helped in improving the F1-score performance for Japanese roads; however, for Indian roads, only the generalizing ability of the models was increased.
3. Visual or qualitative analysis supplements quantitative analysis in accurately assessing the performance across the countries. For example: For mixed damages (like the cases when potholes and Alligator Cracks are inter-mixed for India), there can be multiple possible predictions which are correct, but are marked as incorrect due to mismatching with the information listed in Ground-Truth xmls. Qualitative analysis helps in assessing those cases.
4. Additionally, the visual investigation reveals that the linear joints on the roads are falsely identified as cracks(D00 and D10), increasing the number of false positives. This is in agreement with the observations by the authors in [65]. Similarly, manholes are found to be wrongly identified as potholes (D40). This indicates the requirement of including the categories "D01" and "D11" (longitudinal and transverse joints, respectively) while training the road damage detection models, so that the trained models may differentiate between actual cracks and the crack-like pavement joints. Similarly a category for manholes needs to be included as done in [50].

4.6. Analysis based on mean Average Precision (mAP) of the models

As indicated by the visual analysis, the measure F1-score is unable to accurately identify the improvement in performance (generalizing ability) of models trained using Indian data combined with data from Japan, as compared with models trained using only the Indian data. Apropos, the study considers another quantitative measure - mAP (mean Average Precision) to analyse the performance of the models. mAP has the advantage that it provides a single value for each model to compare its performance with other models. Higher the value of mAP, better is the performance of the model.

The results for Average Precision (AP) and mAP corresponding to different experiments considered in the study are shown in Table 6. The highlighted (bold) entries indicate the highest value of the metric corresponding to test datasets from a single country. For instance, the highlighted entry corresponding to (Experiment E5, AP(D10)) shows the highest value of Average Precision obtained for detecting and classifying Transverse cracks in Japanese data, using all models. This also indicates the superiority of Ja_10k model as compared to the other models considered in the study, for detecting and classifying Transverse cracks in Japanese data.

Similarly, the last column (mAP) shows highlighted entries for the models providing highest mAP for each country. The table shows that for both India and Japan, InJa models provide highest value of mAP (InJa_18k for India and InJa_14k for Japan. This confirms the claims made using visual investigation of the detection results, proving that adding data from other countries improves the generalizing ability and the performance of the models targeted for road damage detection and classification in any country. The future work may address addition of how much foreign data should be considered.

For Czech Republic, the mAP results show Cz_2k performs better than the mixed models JaCz_11k and JaCz_12k. However, the individual values of Average Precision for all Czech models show that the mAP for Cz_2k is highest because of the high values of APs for D20 and D40 corresponding to Cz_2k. The reason for the same is explained in Section 4.3 (target: Czech). Further, the mAP values for JaCz_11k and JaCz_12k correctly illustrates that adding training data from the home country improves the performance of the model.

5. Recommendations and future scope

The following subsections present the proposed recommendations and discuss the scope for extending the work in the future.

5.1. Proposed recommendations

Based on the results and analysis, the study proposes the following recommendations for readers of other countries when one country, (Japan, in this case) releases its data and model:

1. Can they utilize the Japanese model? Would the performance of the model inherited from another country when applied to their country remain the same?
   - Mostly the performance of a model significantly degrades when applied to road images from another country (as evident from the experiments).
   - Readers should perform qualitative analysis by testing the Japanese model on a few images from their country. If the performance is not satisfactory, the readers can follow the steps provided in this work and get a new model for their country using the Japanese model as the baseline.
2. Should they collect images from their country as well? If yes, what effect would these images have on the performance?
   - Yes. The local images help in improving the performance of the model trained on foreign data.
3. Should they mix the images from two countries to train the model or use only the images from their country (pure modeling vs. mixed modeling)?
   - This research recommends the mixing of images collected from different countries due to the following reasons:
     (a) The models can automatically detect and classify road damages quickly once trained using the appropriate data; however, collecting the road images and preparing the corresponding

Table 7
Solutions providing high accuracy for detecting road damages in India, Japan, and the Czech. The models were trained using the dataset proposed in this work and were submitted to Global Road Damage Detection Challenge'2020 [46].

Team name	Data augmentation	Model used	F1-score (Test1)	F1-score (Test2)
IMSC	Synthesized images using Python Augmenter	YOLO-v5 based Ensemble model	0.6748	0.6662
SIS Lab	Used a random crop augmentation algorithm	YOLO-v4 based Ensemble model (30 models)	0.6275	0.6358
DD-VISION	Road segmentation, mixup, CLAHE, RGB shift	A Consistency Filtering Mechanism and cascade R-CNN based Ensemble model	0.6290	0.6219

annotations for Ground-Truth information is a time-taking task. Mixing the images already available from other countries helps in increasing the size of the dataset in lesser time.

(b) It prevents over-fitting and improves the generalizing ability of the model.

(c) Some damage classes can be better represented using data from other countries (For example, D10 for India).

(d) The proposed models are based on transfer learning, and usually, the available models used as a baseline for transfer learning are trained on any general dataset, not necessarily related to road damages. If the road damage data or model is available even though from some other country, using it as the baseline helps the new model learn the required features for automatic road damage detection, faster.

5.2. Future scope

1. This research explores the ways to utilize the data and models available from one country to design road damage detection and classification models for other countries. In the future, the same prototype can be extended to propose a single standardized model that is applicable globally or at least to a set of countries having identical road conditions.
2. Further, this work can be used as a baseline, and the experiments can be repeated by collecting more images from different countries, including India, Japan, and Czech, under different seasonal conditions for better representing each damage class and improving the robustness of the overall detection system for all damage categories.
3. An additional increase in coverage and turnaround time could be achieved by installing the road damage detection system on smartphones, and car vehicle recorders mounted in the vehicles operated by municipalities, such as public transport or waste collection vehicles.
4. Furthermore, public participation through the release of a free smartphone application used by the public for uploading geo-tagged images of poorly performing road sections could provide additional data on the current state of the local road network in several countries.
5. Moreover, only the relative performance of models trained using different datasets were required to carry out this study. Future work can focus on the actual performance of the models leading to designing models with improved accuracy on a similar line.

To explore this aspect of having models with improved performance, we released a pre-print of the current research work [66] and invited solutions from researchers working in the same field through an IEEE Big Data Cup [46]. The Data Cup received huge attention from all over the world [67–78], and in total, 121 teams registered for the competition. The models proposed by the participants were evaluated using two test datasets test1 and test2, comprising 2,631 and 2,664 images, respectively[79]. The top 3 solutions [76–78] solicited by the Big Data Cup are summarized in the Table 7 with more details in [46]. It can be noted that all three solutions involve augmenting the dataset using several techniques and utilize ensemble learning to provide an improved F1-score. In the future, the dataset and the proposed models can be further explored to provide solutions with improved accuracy and better performance in terms of other parameters such as prediction time per image, training time of the models, or the size of the model.

6. Conclusion

The presented study is conducted to address the need of readers of other countries when one other country releases its data and model for automatically detecting and classifying road damages for itself. To achieve the aforementioned objective, the authors proposed a large-scale heterogeneous road damage dataset comprising 26,620 annotated road images collected from multiple countries. Further, the authors explored how the images collected from any country affect the performance of deep learning-based road damage detection and classification model in other countries. The experiments illustrate that an efficient road damage detection model can be developed for a country by mixing the data from another country with the local data. An extensive, comprehensive and rigorous analysis of Japanese data-based models to detect and classify road damages in India and the Czech Republic is presented for the demonstration.

A subset of the proposed dataset has been used for Global Road Damage Detection Challenge (GRDDC'2020) to propose a single model for detecting road damages in three countries - India, Japan, and the Czech. The massive attention(121 registered teams) received by this challenge shows the usefulness of the proposed dataset and the comprehensive analysis carried out in this work for the research community. To support further research, the GRDDC dataset has been made publicly available along with a platform to evaluate the performance of the proposed models.

In essence, the study lays down a foundation for designing a globally applicable standardized model for road damage detection and classification, and is useful for pavement engineers, road authorities, municipalities, and researchers from several countries.

In future, the proposed dataset and model can be extended to support measurement of the size/volume of the detected road damages. External contributions by other researchers are welcome - please contact the authors.

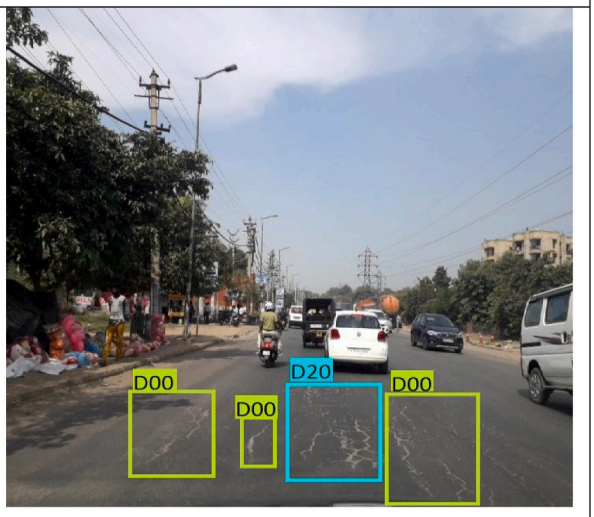
Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

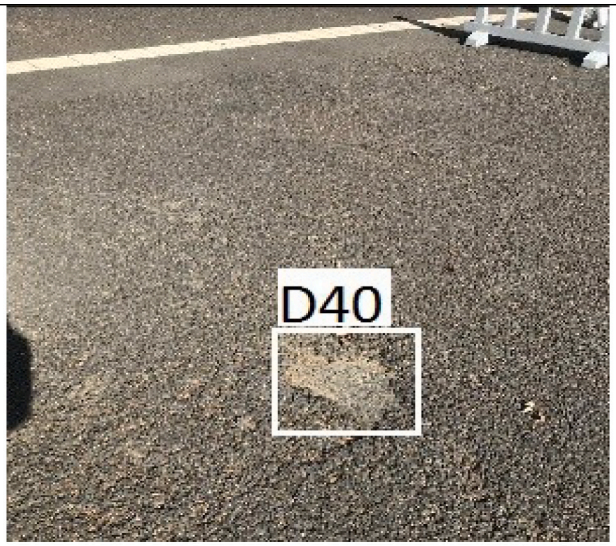
The first author was supported by the Ministry of Education, India, for doctoral work at IIT Roorkee, India, and the Institute of Industrial Science, The University of Tokyo, Japan, for carrying out the research work for this study. Additionally, the authors are grateful to the reviewers for their insightful suggestions to strengthen the quality of the manuscript.

Appendix. Ground truth for the images used for visual analysis

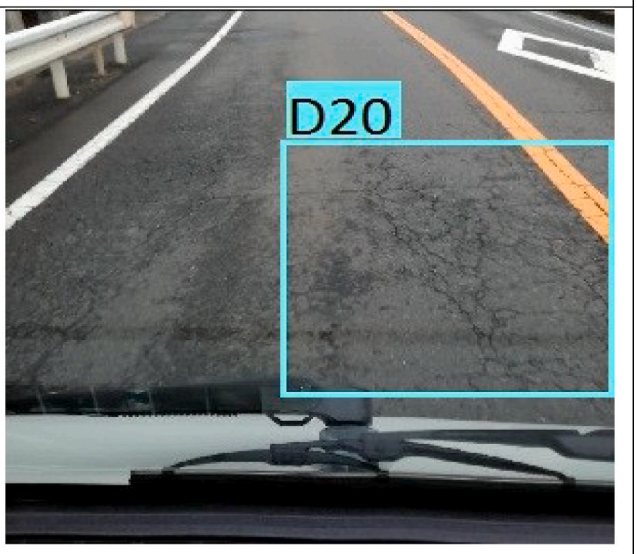
	Road Image	Ground Truth Visualization
1.		
2.		
3.		

1449

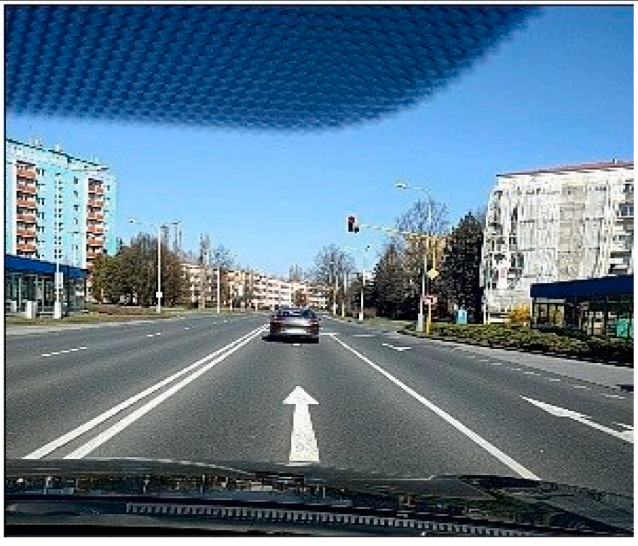
4.

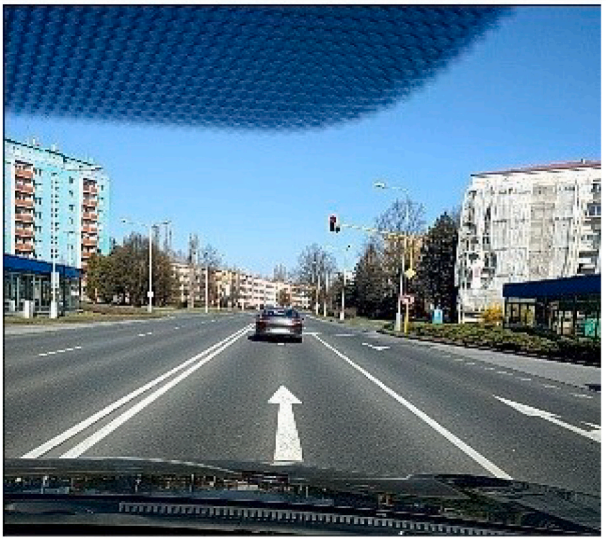


5.



6.





. (*continued*).

7.

8.

. (*continued*).

References

[1] T.R. Miller, E. Zaloshnja, On a Crash Course: The Dangers and Health Costs of Deficient Roadways, Technical Report, The Transportation Construction Coalition, United States, 2009. Retrieved from: http://www.infrastructureusa.org/on-a-crash-course-the-dangers-and-health-costs-of-deficient-roadways/, Last Accessed: 2021-07-26.

[2] M. Rojo, H. Gonzalo-Orden, A. Linares, et al., Impact of a lower conservation budget on road safety indices, J. Adv. Transp. 2018 (2018), https://doi.org/10.1155/2018/9570465.

[3] E. Zalama, J. Gómez-García-Bermejo, R. Medina, J. Llamas, Road crack detection using visual features extracted by Gabor filters, Comput. Aided Civ. Infrastruct. Eng. 29 (2014) 342–358, https://doi.org/10.1111/mice.12042.

[4] L.M. Pierce, G. McGovern, K.A. Zimmerman, Practical Guide for Quality Management of Pavement Condition Data Collection, Technical Report, United States Department of Transportation, Federal Highway Administration, 2013. Retrieved from: http://www.fhwa.dot.gov/pavement/management/qm/data_qm_guide.pdf, Last Accessed: 2021-07-22.

[5] K. McGhee, Automated Pavement Distress Collection Techniques, National Cooperative Highway Research Program, NCHRP Synthesis 334 Report, Transportation Research Board, Washington DC, 2004, https://doi.org/10.17226/23348.

[6] M. Eisenbach, R. Stricker, D. Seichter, K. Amende, K. Debes, M. Sesselmann, D. Ebersbach, U. Stoeckert, H.-M. Gross, How to get pavement distress detection ready for deep learning? A systematic approach, in: International Joint Conference on Neural Networks (IJCNN), 2017, pp. 2039–2047, https://doi.org/10.1109/IJCNN.2017.7966101.

[7] D. Zhang, Q. Zou, H. Lin, X. Xu, L. He, R. Gui, Q. Li, Automatic pavement defect detection using 3D laser profiling technology, Autom. Constr. 96 (2018) 350–365, https://doi.org/10.1016/j.autcon.2018.09.019.

[8] J. Guan, X. Yang, L. Ding, X. Cheng, V.C. Lee, C. Jin, Automated pixel-level pavement distress detection based on stereo vision and deep learning, Autom. Constr. 129 (2021) 103788, https://doi.org/10.1016/j.autcon.2021.103788.

[9] S.C. Radopoulou, I. Brilakis, Automated detection of multiple pavement defects, J. Comput. Civ. Eng. 31 (2017) 04016057, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000623.

[10] C. Mertz, S. Varadharajan, S. Jose, K. Sharma, L. Wander, J. Wang, City-wide road distress monitoring with smartphones, in: Proceedings of ITS World Congress, 2014, pp. 1–9. Retrieved from: https://www.ri.cmu.edu/pub_files/2014/9/road_monitor_mertz_final.pdf, Last Accessed: 2021-07-28.

[11] D.A. Casas-Avellaneda, J.F. López-Parra, Detection and localization of potholes in roadways using smartphones, Dyna 83 (2016) 156–162, https://doi.org/10.15446/dyna.v83n195.44919.

[12] H. Maeda, Y. Sekimoto, T. Seto, T. Kashiyama, H. Omata, Road damage detection and classification using deep neural networks with smartphone images, Comput. Aided Civ. Infrastruct. Eng. 33 (2018) 1127–1141, https://doi.org/10.1111/mice.12387.

[13] I. Goodfellow, Y. Bengio, A. Courville, Deep Learning, MIT press, 2016. http://www.deeplearningbook.org. , Last Accessed: 2021-07-28.

[14] A. Krizhevsky, I. Sutskever, G.E. Hinton, Imagenet classification with deep convolutional neural networks, Adv. Neural Inf. Process. Syst. 25 (2012) 1097–1105, https://doi.org/10.1145/3065386.

[15] R. Girshick, J. Donahue, T. Darrell, J. Malik, Rich feature hierarchies for accurate object detection and semantic segmentation, in: Proceedings of the IEEE

Conference on Computer Vision and Pattern Recognition, 2014, pp. 580–587, https://doi.org/10.1109/CVPR.2014.81.
[16] R. Girshick, Fast R-CNN, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 1440–1448, https://doi.org/10.1109/ICCV.2015.169.
[17] S. Ren, K. He, R. Girshick, J. Sun, Faster R-CNN: towards real-time object detection with region proposal networks. Advances in Neural Information Processing Systems, 2015, pp. 91–99, https://doi.org/10.1109/TPAMI.2016.2577031.
[18] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778, https://doi.org/10.1109/CVPR.2016.90.
[19] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, A.C. Berg, SSD: single shot multibox detector, in: European Conference on Computer Vision, Springer, 2016, pp. 21–37, https://doi.org/10.1007/978-3-319-46448-0_2.
[20] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2818–2826, https://doi.org/10.1109/CVPR.2016.308.
[21] J. Huang, V. Rathod, C. Sun, M. Zhu, A. Korattikara, A. Fathi, I. Fischer, Z. Wojna, Y. Song, S. Guadarrama, et al., Speed/accuracy trade-offs for modern convolutional object detectors, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 7310–7311, https://doi.org/10.1109/CVPR.2017.351.
[22] A.G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, H. Adam, Mobilenets: Efficient Convolutional Neural Networks for Mobile Vision Applications, 2017 arXiv preprint arXiv:1704.04861.
[23] S. Biçici, M. Zeybek, An approach for the automated extraction of road surface distress from a UAV-derived point cloud, Autom. Constr. 122 (2021) 103475, https://doi.org/10.1016/j.autcon.2020.103475.
[24] R. Stricker, M. Eisenbach, M. Sesselmann, K. Debes, H.-M. Gross, Improving visual road condition assessment by extensive experiments on the extended gaps dataset, in: International Joint Conference on Neural Networks (IJCNN), 2019, pp. 1–8, https://doi.org/10.1109/IJCNN.2019.8852257.
[25] A. Zhang, K.C. Wang, B. Li, E. Yang, X. Dai, Y. Peng, Y. Fei, Y. Liu, J.Q. Li, C. Chen, Automated pixel-level pavement crack detection on 3D asphalt surfaces using a deep-learning network, Comput. Aided Civ. Infrastruct. Eng. 32 (2017) 805–819, https://doi.org/10.1111/mice.12297.
[26] L. Zhang, F. Yang, Y.D. Zhang, Y.J. Zhu, Road crack detection using deep convolutional neural network, in: 2016 IEEE International Conference on Image Processing (ICIP), IEEE, 2016, pp. 3708–3712, https://doi.org/10.1109/ICIP.2016.7533052.
[27] W.R.L.d. Silva, D.S.d. Lucena, Concrete cracks detection based on deep learning image classification, in: Multidisciplinary Digital Publishing Institute Proceedings, volume 2, 2018, p. 489, https://doi.org/10.3390/ICEM18-05387.
[28] S. Anand, S. Gupta, V. Darbari, S. Kohli, Crack-pot: autonomous road crack and pothole detection, in: 2018 Digital Image Computing: Techniques and Applications (DICTA), IEEE, 2018, pp. 1–6, https://doi.org/10.1109/DICTA.2018.8615819.
[29] Z. Fan, Y. Wu, J. Lu, W. Li, Automatic Pavement Crack Detection Based on Structured Prediction with the Convolutional Neural Network, 2018 arXiv preprint arXiv:1802.02208.
[30] H. Nhat-Duc, Q.-L. Nguyen, V.-D. Tran, Automatic recognition of asphalt pavement cracks using metaheuristic optimized edge detection algorithms and convolution neural network, Autom. Constr. 94 (2018) 203–213, https://doi.org/10.1016/j.autcon.2018.07.008.
[31] Y.O. Ouma, M. Hahn, Pothole detection on asphalt pavements from 2D-colour pothole images using fuzzy c-means clustering and morphological reconstruction, Autom. Constr. 83 (2017) 196–211, https://doi.org/10.1016/j.autcon.2017.08.017.
[32] H. Oliveira, P.L. Correia, CrackIT—an image processing toolbox for crack detection and characterization, in: 2014 IEEE International Conference on Image Processing (ICIP), IEEE, 2014, pp. 798–802, https://doi.org/10.1109/ICIP.2014.7025160.
[33] Y. Shi, L. Cui, Z. Qi, F. Meng, Z. Chen, Automatic road crack detection using random structured forests, IEEE Trans. Intell. Transp. Syst. 17 (2016) 3434–3445, https://doi.org/10.1109/TITS.2016.2552248.
[34] L. Cui, Z. Qi, Z. Chen, F. Meng, Y. Shi, Pavement distress detection using random decision forests, in: International Conference on Data Science, Springer, 2015, pp. 95–102, https://doi.org/10.1007/978-3-319-24474-7_14.
[35] Q. Zou, Y. Cao, Q. Li, Q. Mao, S. Wang, CrackTree: automatic crack detection from pavement images, Pattern Recogn. Lett. 33 (2012) 227–238, https://doi.org/10.1016/j.patrec.2011.11.004.
[36] X. Weng, Y. Huang, W. Wang, Segment-based pavement crack quantification, Autom. Constr. 105 (2019) 102819, https://doi.org/10.1016/j.autcon.2019.04.014.
[37] S. Dorafshan, R.J. Thomas, M. Maguire, SDNET2018: an annotated image dataset for non-contact concrete crack detection using deep convolutional neural networks, Data Brief. 21 (2018) 1664–1668, https://doi.org/10.1016/j.dib.2018.11.015.
[38] L. Li, L. Sun, G. Ning, S. Tan, Automatic pavement crack recognition based on BP neural network, PROMET-Traffic Transp. 26 (2014) 11–22, https://doi.org/10.7307/ptt.v26i1.1477.
[39] F. Yang, L. Zhang, S. Yu, D. Prokhorov, X. Mei, H. Ling, Feature pyramid and hierarchical boosting network for pavement crack detection, IEEE Trans. Intell. Transp. Syst. 21 (2020) 1525–1535, https://doi.org/10.1109/TITS.2019.2910595.
[40] Q. Mei, M. Gül, A cost effective solution for pavement crack inspection using cameras and deep neural networks, Constr. Build. Mater. 256 (2020) 119397, https://doi.org/10.1016/j.conbuildmat.2020.119397.
[41] H. Majidifard, P. Jin, Y. Adu-Gyamfi, W.G. Buttlar, Pavement image datasets: a new benchmark dataset to classify and densify pavement distresses, Transp. Res. Rec. 2674 (2020) 328–339, https://doi.org/10.1177/0361198120907283.
[42] A. Angulo, J.A. Vega-Fernández, L.M. Aguilar-Lobo, S. Natraj, G. Ochoa-Ruiz, Road damage detection acquisition system based on deep neural networks for physical asset management, in: Mexican International Conference on Artificial Intelligence, Springer, 2019, pp. 3–14, https://doi.org/10.1007/978-3-030-33749-0_1.
[43] E. Karaaslan, U. Bagci, F.N. Catbas, Attention-guided analysis of infrastructure damage with semi-supervised deep learning, Autom. Constr. 125 (2021) 103634, https://doi.org/10.1016/j.autcon.2021.103634.
[44] V.M. Souza, R. Giusti, A.J. Batista, Asfault: a low-cost system to evaluate pavement conditions in real-time using smartphones and machine learning, Pervasive Mobile Comput. 51 (2018) 121–137, https://doi.org/10.1016/j.pmcj.2018.10.008.
[45] G. Singh, D. Bansal, S. Sofat, N. Aggarwal, Smart patrolling: an efficient road surface monitoring using smartphone sensors and crowdsourcing, Pervasive Mobile Comput. 40 (2017) 71–88, https://doi.org/10.1016/j.pmcj.2017.06.002.
[46] D. Arya, H. Maeda, S. Kumar Ghosh, D. Toshniwal, H. Omata, T. Kashiyama, Y. Sekimoto, Global road damage detection: state-of-the-art solutions, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5533–5539, https://doi.org/10.1109/BigData50022.2020.9377790.
[47] M. Abdellatif, H. Peel, A.G. Cohn, R. Fuentes, Combining block-based and pixel-based approaches to improve crack detection and localisation, Autom. Constr. 122 (2021) 103492, https://doi.org/10.1016/j.autcon.2020.103492.
[48] S. Xie, Z. Tu, Holistically-nested edge detection, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 1395–1403, https://doi.org/10.1007/s11263-017-1004-z.
[49] Y. Liu, M.-M. Cheng, X. Hu, K. Wang, X. Bai, Richer convolutional features for edge detection, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 3000–3009, https://doi.org/10.1109/tpami.2018.2878849.
[50] H. Maeda, T. Kashiyama, Y. Sekimoto, T. Seto, H. Omata, Generative adversarial network for road damage detection, Comput. Aided Civ. Infrastruct. Eng. (2020), https://doi.org/10.1111/mice.12561.
[51] A. Alfarrarjeh, D. Trivedi, S.H. Kim, C. Shahabi, A deep learning approach for road damage detection from smartphone images, in: 2018 IEEE International Conference on Big Data (Big Data), IEEE, 2018, pp. 5201–5204, https://doi.org/10.1109/BigData.2018.8621899.
[52] F. Kluger, C. Reinders, K. Raetz, P. Schelske, B. Wandt, H. Ackermann, B. Rosenhahn, Region-based cycle-consistent data augmentation for object detection, in: 2018 IEEE International Conference on Big Data (Big Data), IEEE, 2018, pp. 5205–5211, https://doi.org/10.1109/BigData.2018.8622318.
[53] Y.J. Wang, M. Ding, S. Kan, S. Zhang, C. Lu, Deep proposal and detection networks for road damage detection and classification, in: 2018 IEEE International Conference on Big Data (Big Data), IEEE, 2018, pp. 5224–5227, https://doi.org/10.1109/BigData.2018.8622599.
[54] W. Wang, B. Wu, S. Yang, Z. Wang, Road damage detection and classification with faster R-CNN, in: 2018 IEEE International Conference on Big Data (Big Data), IEEE, 2018, pp. 5220–5223, https://doi.org/10.1109/BigData.2018.8622354.
[55] R. Roberts, G. Giancontieri, L. Inzerillo, G. Di Mino, Towards low-cost pavement condition health monitoring and analysis using deep learning, Appl. Sci. 10 (2020) 319, https://doi.org/10.3390/app10010319.
[56] Y. Du, N. Pan, Z. Xu, F. Deng, Y. Shen, H. Kang, Pavement distress detection and classification based on YOLO network, Int. J. Pavement Eng. (2020) 1–14, https://doi.org/10.1080/10298436.2020.1714047.
[57] S. Patra, A.I. Middya, S. Roy, PotSpot: participatory sensing based monitoring system for pothole detection using deep learning, Multimed. Tools Appl. (2021) 1–25, https://doi.org/10.1007/s11042-021-10874-4.
[58] D. Arya, H. Maeda, S.K. Ghosh, D. Toshniwal, Y. Sekimoto, RDD2020: an annotated image dataset for automatic road damage detection using deep learning, Data Brief. (2021) 107133, https://doi.org/10.1016/j.dib.2021.107133.
[59] Maintenance Guidebook for Road Pavements, 2013 edition, Technical Report, Japan Road Association, Japan, 2013. Retrieved from: http://www.road.or.jp/english/publication/index.html, Last Accessed: 2021-07-28.
[60] Catalog of Deficiencies in Flexible Pavements, Technical Report TP-82, Czech Ministry of Transport, Road Transport Section, Czech Republic, 2009. Retrieved from: http://www.pjpk.cz/technicke-podminky-tp/, Last Accessed: 2021-07-30.
[61] IRC:82, Code of Practice for Maintenance of Bituminous Road Surfaces, Technical Report, Indian Road Congress, New Delhi, India, 2015. Retrieved from: http://www.irc.nic.in/, Last Accessed: 2021-07-28.
[62] M. Everingham, L. Van Gool, C.K. Williams, J. Winn, A. Zisserman, The Pascal Visual Object Classes (VOC) challenge, Int. J. Comput. Vis. 88 (2010) 303–338, https://doi.org/10.1007/s11263-009-0275-4.
[63] Intel, Pedestrian Detection Using Tensorflow on Intel Architecture (White Paper), Intel Ai Builders, 2018. Retrieved from: https://builders.intel.com/docs/aibuilders/pedestrian-detection-using-tensorflow-on-intel-architecture.pdf, Last Accessed: 2021-07-28.
[64] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C. L. Zitnick, Microsoft COCO: common objects in context, in: European Conference on Computer Vision, Springer, 2014, pp. 740–755, https://doi.org/10.1007/978-3-319-10602-1_48.
[65] N. Kargah-Ostadi, A. Nazef, J. Daleiden, Y. Zhou, Evaluation framework for automated pavement distress identification and quantification applications, Transp. Res. Rec. 2639 (2017) 46–54, https://doi.org/10.3141/2639-06.
[66] D. Arya, H. Maeda, S.K. Ghosh, D. Toshniwal, A. Mraz, T. Kashiyama, Y. Sekimoto, Transfer Learning-Based Road Damage Detection for Multiple Countries, 2020 arXiv preprint arXiv:2008.13101.

[67] V. Pham, C. Pham, T. Dang, Road damage detection and classification with Detectron2 and faster R-CNN, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5592–5601, https://doi.org/10.1109/BigData50022.2020.9378027.
[68] V. Mandal, A.R. Mussah, Y. Adu-Gyamfi, Deep learning frameworks for pavement distress classification: a comparative analysis, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5577–5583, https://doi.org/10.1109/BigData50022.2020.9378047.
[69] S. Naddaf-Sh, M.-M. Naddaf-Sh, A.R. Kashani, H. Zargarzadeh, An efficient and scalable deep learning approach for road damage detection, in: 2020 IEEE International Conference on Big Data (Big Data), IEEE, 2020, pp. 5602–5608, https://doi.org/10.1109/BigData50022.2020.9377751.
[70] T. Hascoet, Y. Zhang, A. Persch, R. Takashima, T. Takiguchi, Y. Ariki, FasterRCNN monitoring of road damages: competition and deployment, in: 2020 IEEE International Conference on Big Data (Big Data), IEEE, 2020, pp. 5545–5552, https://doi.org/10.1109/BigData50022.2020.9377871.
[71] F. Kortmann, K. Talits, P. Fassmeyer, A. Warnecke, N. Meier, J. Heger, P. Drews, B. Funk, Detecting various road damage types in global countries utilizing faster R-CNN, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5563–5571, https://doi.org/10.1109/BigData50022.2020.9378245.
[72] Y. Liu, X. Zhang, B. Zhang, Z. Chen, Deep network for road damage detection, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5572–5576, https://doi.org/10.1109/BigData50022.2020.9377991.
[73] R. Vishwakarma, R. Vennelakanti, CNN model tuning for global road damage detection, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5609–5615, https://doi.org/10.1109/BigData50022.2020.9377902.
[74] X. Zhang, X. Xia, N. Li, M. Lin, J. Song, N. Ding, Exploring the tricks for road damage detection with a one-stage detector, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5616–5621, https://doi.org/10.1109/BigData50022.2020.9377923.
[75] D. Jeong, Road damage detection using YOLO with smartphone images, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5559–5562, https://doi.org/10.1109/BigData50022.2020.9377847.
[76] V. Hegde, D. Trivedi, A. Alfarrarjeh, A. Deepak, S. Ho Kim, C. Shahabi, Yet another deep learning approach for road damage detection using ensemble learning, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5553–5558, https://doi.org/10.1109/BigData50022.2020.9377833.
[77] K. Doshi, Y. Yilmaz, Road damage detection using deep ensemble learning, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5540–5544, https://doi.org/10.1109/BigData50022.2020.9377774.
[78] Z. Pei, R. Lin, X. Zhang, H. Shen, J. Tang, Y. Yang, CFM: a consistency filtering mechanism for road damage detection, in: 2020 IEEE International Conference on Big Data (Big Data), 2020, pp. 5584–5591, https://doi.org/10.1109/BigData50022.2020.9377911.
[79] , Dataset v1, 2021. RDD2020: an image dataset for smartphone-based road damage detection and classification. (Accessed 31 August 2021).